🌍 Language Translation Tool

A modern web-based Language Translation Tool that allows users to translate text between multiple languages using the Google Translate API. The application provides a clean user interface along with advanced features such as voice input, text-to-speech, translation history, dark mode, and language swapping.

📌 Project Overview

The Language Translation Tool is designed to help users translate text quickly and efficiently between different languages. It leverages Google Translate services to provide accurate translations while offering an intuitive and responsive user experience.

This project demonstrates API integration, DOM manipulation, Local Storage usage, speech recognition, and responsive web design using HTML, CSS, and JavaScript.

🚀 Features

- 🌐 Translate text into multiple languages
- 🔄 Auto language detection
- ⇄ Swap source and target languages
- 🎤 Voice input using Speech Recognition
- 🔊 Text-to-Speech for translated output
- 📋 Copy translated text to clipboard
- 🌙 Dark Mode / Light Mode
- 📊 Character Counter
- 📝 Translation History using Local Storage
- 📱 Responsive Design for mobile and desktop devices

🛠️ Technologies Used

Technology	Purpose
HTML5	Structure of the application
CSS3	Styling and responsive design
JavaScript (ES6)	Application functionality
Google Translate API	Language translation
Web Speech API	Voice recognition and text-to-speech

Technology	Purpose
Local Storage	Saving translation history and theme preferences

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📂 Project Structure

Language-Translator/

├── index.html

├── style.css

├── script.js

└── README.md

⚙️ Installation & Usage

Step 1: Clone the Repository

git clone https://github.com/your-username/language-translator.git

Step 2: Open Project Folder

cd language-translator

Step 3: Run the Project

Open the index.html file in your browser

OR

Use VS Code Live Server for better performance.

📖 How to Use

1. Enter text in the input box.
2. Select the source language or use Auto Detect.
3. Select the target language.
4. Click the Translate button.
5. View the translated text instantly.

6. Use:

7. 🎤 Voice button for speech input

8. 🔊 Speak button to hear the translation
9. 📋 Copy button to copy translated text
10. ⇄ Swap button to exchange languages

📸 Screenshots

Add screenshots of your project here.

Example:

- Home Page
- Translation Output
- Dark Mode Interface
- Voice Input Feature

🎯 Learning Outcomes

Through this project, I learned:

- API Integration using Fetch API
- Asynchronous JavaScript (async/await)
- DOM Manipulation
- Event Handling
- Local Storage Implementation
- Responsive UI Design
- Speech Recognition and Text-to-Speech APIs

🔮 Future Enhancements

- Support for more languages
- Translation download as PDF
- OCR Translation from Images
- Real-time translation while typing
- User authentication and saved history
- Translation statistics dashboard

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📄 License